LAPORAN PRAKTIKUM
WORKSHOP SISTEM INFORMASI WEB SERVER
(TIF330805)
SEMESTER III



Acara 20
API Authentication dan Error Handling

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A. Tujuan Pengujian

Pengujian dilakukan untuk memastikan endpoint API mahasiswa dapat menjalankan operasi GET, POST, PUT, dan DELETE dengan benar serta menerapkan autentikasi JSON Web Token (JWT) pada endpoint yang memerlukan akses terproteksi.

Pengujian dilakukan menggunakan **curl.exe** melalui terminal PowerShell sebagai alternatif dari Postman.

1. Metode GET

```
$base = "http://localhost/BKPM/Acara20/si-
akademik/api/mahasiswa"
curl.exe -i $base
```

2. Metode POST

```
$body = '{"nim":"2024004","nama":"Budi
Uji","email":"budiuji@mail.com","jurusan":"Teknik
Informatika"}'
[System.IO.File]::WriteAllText(
    "$PWD\body.json",
    $body,
    [System.Text.UTF8Encoding]::new($false)
)
curl.exe -i -X POST $base `
  -H "Content-Type: application/json" `
  --data-binary "@body.json"
```

3. Metode PUT

```
$body = '{"nim":"2024001","nama":"Budi Santoso
Updated","email":"budi@mail.com","jurusan":"Sistem
Informasi"}'
[System.IO.File]::WriteAllText(
```

```
    "$PWD\body.json",
    $body,
    [System.Text.UTF8Encoding]::new($false)
)
curl.exe -i -X PUT "${base}?id=1" `
  -H "Content-Type: application/json" `
  --data-binary "@body.json"
```

4. Metode DELETE

```
curl.exe -i -X DELETE "${base}?id=1"
```

- **$base** digunakan untuk menyimpan URL endpoint API mahasiswa agar dapat digunakan kembali pada perintah berikutnya.
- **http://localhost** menunjukkan bahwa aplikasi dijalankan pada komputer lokal.
- **/BKPM/Acara20/si-akademik/api/mahasiswa** merupakan alamat endpoint yang menangani data mahasiswa
- **curl.exe** digunakan untuk mengirim permintaan HTTP ke server.
- **-X POST** menentukan bahwa permintaan HTTP menggunakan metode POST untuk mengirim data baru ke API.
- **-X PUT** menentukan bahwa permintaan HTTP digunakan untuk memperbarui data.
- **-X DELETE** menentukan bahwa permintaan HTTP digunakan untuk menghapus data.
- **-i** digunakan untuk menampilkan header respons HTTP beserta isi respons dari server.
- **-H "Content-Type: application/json**" memberi tahu server bahwa isi permintaan menggunakan format JSON.
- **--data-binary "@body.json"** mengirim isi file body.json sebagai badan permintaan (request body).

B. Hasil Pengujian

```
PS C:\laragon\www\BKPM> cd Acara20
PS C:\laragon\www\BKPM\Acara20> $base_url = "http://localhost/BKPM/Acara20/si-akademik/public"
>> $login_url = "$base_url/login"
>> $mahasiswa_url = "$base_url/mahasiswa"
PS C:\laragon\www\BKPM\Acara20>
```

```
PS C:\laragon\www\BKPM\Acara20> $body = '{"username":"andi_test","password":"Rahasia123!"}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
>>
>> curl.exe -i -X POST "$login_url" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json"
HTTP/1.1 200 OK
Date: Wed, 21 Oct 2026 04:28:15 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 359
Content-Type: application/json; charset=utf-8

{
    "success": true,
    "message": "Login berhasil.",
    "data": {
        "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6MSwicm9sZSI6InVzZXIiLCJ1c2VybmFtZSI6ImFuZGlfdGVzdCIsImV4cCI6MTc5MjU1NzE5NX0.A5QYSK2k9cnFiXGc7oSMCBZ0
g_80p9DfHvG5q1abApE",
        "role": "user",
        "expires_in": 300
    },
    "timestamp": "2026-10-21T04:28:15+00:00"
}
```

1. Login menggunakan akun dari regist di acara 19

```
PS C:\laragon\www\BKPM\Acara20> $response = curl.exe -s -X POST "$login_url" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json" |
>>   ConvertFrom-Json
>>
>> $response | ConvertTo-Json -Depth 10
>>
>> $token = $response.data.token
{
    "success":  true,
    "message":  "Login berhasil.",
    "data":  {
                 "token":  "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6MSwicm9sZSI6InVzZXIiLCJ1c2VybmFtZSI6ImFuZGlfdGVzdCIsImV4cCI6MTc5MjU1NzIzNH0.tllawtYkHOPM-B
AEeZZ91Ooaj0-9qYSNey_GKIT6zkQ",
                 "role":  "user",
                 "expires_in":  300
             },
    "timestamp":  "2026-10-21T04:28:54+00:00"
}
```

2. Menyimpan token otomatis agar tidak di salinn berulang ulang

```
PS C:\laragon\www\BKPM\Acara20> $body = '{"nim":"220100199","nama":"Uji JWT","prodi":"Teknik Informatika"}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
>>
>> curl.exe -i -X POST "$mahasiswa_url" `
>>   -H "Content-Type: application/json" `
>>   -H "Authorization: Bearer $token" `
>>   --data-binary "@body.json"
HTTP/1.1 201 Created
Date: Wed, 21 Oct 2026 04:29:37 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 153
Content-Type: application/json; charset=utf-8

{
    "success": true,
    "message": "Data berhasil ditambahkan.",
    "data": {
        "id": "3"
    },
    "timestamp": "2026-10-21T04:29:37+00:00"
}
```

3. Menambahkan data mahasiswa menggunakan method HTTP POST

```
PS C:\laragon\www\BKPM\Acara20> $body = '{"nim":"202400101","nama":"Budi JWT Updated","prodi":"Sistem Informasi"}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
>>
>> curl.exe -i -X PUT "${mahasiswa_url}?id=1" `
>>   -H "Content-Type: application/json" `
>>   -H "Authorization: Bearer $token" `
>>   --data-binary "@body.json"
HTTP/1.1 200 OK
Date: Wed, 21 Oct 2026 04:33:04 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 150
Content-Type: application/json; charset=utf-8

{
    "success": true,
    "message": "Data berhasil diperbarui.",
    "data": {
        "id": 1
    },
    "timestamp": "2026-10-21T04:33:04+00:00"
}
PS C:\laragon\www\BKPM\Acara20>
```

4. Menambahkan data mahasiswa menggunakan method HTTP PUT

```
PS C:\laragon\www\BKPM\Acara20> curl.exe -i -X DELETE "${mahasiswa_url}?id=1" `
>>   -H "Authorization: Bearer $token"
HTTP/1.1 200 OK
Date: Wed, 21 Oct 2026 04:36:16 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 147
Content-Type: application/json; charset=utf-8

{
    "success": true,
    "message": "Data berhasil dihapus.",
    "data": {
        "id": 1
    },
    "timestamp": "2026-10-21T04:36:16+00:00"
}
```

5. Menambahkan data mahasiswa menggunakan method HTTP DELETE

```
PS C:\laragon\www\BKPM\Acara20> curl.exe -i -X POST "$mahasiswa_url" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json"
HTTP/1.1 401 Unauthorized
Date: Wed, 21 Oct 2026 04:36:54 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 173
Content-Type: application/json; charset=utf-8

{
    "success": false,
    "message": "Token tidak ditemukan. Kirim header Authorization: Bearer <token>.",
    "data": null,
    "timestamp": "2026-10-21T04:36:54+00:00"
}
```

6. Menambahkan data mahasiswa menggunakan method HTTP POST Tanpa Token

```
PS C:\laragon\www\BKPM\Acara20> $badToken = $token + "x"
PS C:\laragon\www\BKPM\Acara20> curl.exe -i -X PUT "${mahasiswa_url}?id=2" `
>>   -H "Content-Type: application/json" `
>>   -H "Authorization: Bearer $badToken" `
>>   --data-binary "@body.json"
HTTP/1.1 401 Unauthorized
Date: Wed, 21 Oct 2026 04:38:07 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 142
Content-Type: application/json; charset=utf-8

{
    "success": false,
    "message": "Token tidak valid atau kedaluwarsa.",
    "data": null,
    "timestamp": "2026-10-21T04:38:07+00:00"
}
```

7. Pengujian update data mahasiswa menggunakan HTTP PUT dengan token yang diubah